



Dr. MANISH KUMAR

Assistant Professor

Department of Electrical Engineering

School of Engineering and Technology

Central University of Haryana.

Phone Number - +919255140623

ORCID	https://orcid.org/0000-0002-4827-6642
Google Scholar id	https://scholar.google.com/citations?user=cnV7C-4AAAAJ&hl=en
Email id	khanagwal.manish@gmail.com , manish.kumar@cuh.ac.in
Area of Specialization	• Renewable Energy Sources • Electrical Vehicle • Optimization Techniques • Deregulated Electricity Markets • Congestion Management of Transmission and Distribution System • Nodal price Management of Transmission and Distribution System
Academic Education	Ph.D – Electrical Engineering, NIT Kurushetra M.Tech -- Electrical Engineering (Power System), NIT Kurushetra B.E -- Instrumentation Engineering, SLIET Longowal, Punjab Diploma – Electrical Engineering, Govt. Polytechnic Sirsa
Experience (Total=15+)	• 8+ Assistant Professor in Electrical Engg. Deptt. SOE&T, CUH • Five year four month research as a Senior Research Fellow in NIT Kurukshetra. • Assistant Professor in Doon Vally Engg. College, Karnal.2011. • One year teaching as an Lecture in CDLM Engg.College Paniwala mota Sirsa.
Administrative Experience: Contributions to Academic Enrichment	• OSD Vice- Chancellor (Infra Maintance) in CUH • Executive Engineer (Electrical) in CUH • M.tech Course Co-Ordinator • Member of School Board (SOET) • Member of BOS • Member of DRC • Coordinator of M.tech Admission Committee • Coordinator of B.tech Admission Committee • Member of Ph.D Admission Committee • Member of Proctorial Duty in 10th convocation 2024 • Member of some purchase committee in CUH • Lab In-charge of Machine and M.Tech lab • Nodal Officer (Nasha Mukht Bhart Abhyan) • Member of Security, Parking and Discipline Committee in the event of Inauguration Ceremony of one wing of Boys Hostel;

	unveiling of Statue of Dr. B.R Ambedkar and laying the Foundation Stone of Water Treatment Plant .
Research Supervision	Ph D =2 , M. Tech = 04
Research Contribution	
Patent (Total= 06)	1. Muralidhar Nayak Bhukya and Manish Kumar, et al, “Machine Learning based Cyber security system for Smart Grid”, application no-202011051586A, issue no-9/2020, Date of Publication:04/12/2020. (published) 2. Muralidhar Nayak Bhukya and Manish Kumar, et al, “3D Printer”, application no-343104-001, issue no-27/2021, Date of Publication: 02/07/2021. (Design accepted and published) 3. Manish Kumar and Akshat Kant, et al, “Isothermal Electric Stove “application no-353588-001, issue no- 03/2023, patent certificate no-126390. Date of Publication 24/11/2021. (Design Granted and Published) 4. Manish Kumar and Sangeeta, et al, “Facial Emotions based online exam cheat detection”, application no-202411015665A, issue no-10/2024, Date of Publication 08/03/2024. 5. Manish Kumar and Dr.Babita Rani, et al, “Intelligent Classroom Management System for Enhanced Learning” ,application no-202411017159A, issue no- 10/2024, Date of Publication 12/04/2024. 6. Manish Kumar and Sangeeta, et al, “Renewable Energy Powered Poles to detect noise pollution”, Design no- 450176-001, Design Granted on 03/03/2025.
Publication (Journals =27, Book chapter = 2, Conference = 16)	Published in Journals(SCI/SCIE/Scopus) 1. Akshat Kant, Manish Kumar & Sathans Sihag, “Enhanced battery management system for li-ion batteries using hybrid machine learning model for accurate state of health prediction”, International Journal of Ambient Energy, Vol.46,No.1, Page 1-9 , 2025. (Taylor & Francis, Scopus Index with impact factor 6.3) DOI: 10.1080/01430750.2025.2503276 2. Akshat Kant, Manish Kumar & Sathans Sihag, “Prediction of electric vehicle battery state of health estimation using a hybrid deep learning mechanism”, International Journal of Green Energy, Vol. 22, Issue.10, Page no.1-15, 2025. (Taylor & Francis, SCIE Index with impact factor 3.4) DOI: 10.1080/15435075.2024.2448301 3. Parveen Kumar, Manish Kumar, Ajay Kumar Bansal, “Automatic Fault Detection of Photovoltaic Modules Using Recurrent Neural Network”, Russian Electrical Engineering 95 (4), 321-334. (Springer) DOI: 10.3103/S1068371224700330 4. Parveen Kumar, Manish Kumar, and Ajay Kumar Bansal, “Photovoltaic (PV) Parameter Extraction using a Hybrid Algorithm based on Spotted Hyena-Ant Lion Optimization,” International Journal

on Electrical Engineering and Informatics, Vol. 15, No. 4, December 2023. (Scopus Index) <https://www.ijeei.org/docs-121546105465a3db9c88312.pdf>

5. Parveen Kumar, **Manish Kumar**, Ajay Kumar using Spotted Hyena and Quadratic Approximation based Hybrid Algorithm under Partially Bansal, "GMPP Tracking of Solar PV system shaded conditions", IEEE ACCESS, 2023. DOI: 10.1109/ACCESS. 2023. 3272554 (SCI index)
6. Manish Kumar, K. S. Sandhu, Ashwani Kumar, "Wind speed variation impact on transmission loss reduction in electricity market", Procedia computer science, vol.70, pp-526-537, 2015. <https://doi.org/10.1016/j.procs.2015.10.095>
7. Manish Kumar, Ashwani Kumar, K.S. Sandhu, "Optimal Location of WT based Distributed Generation in Pool based Electricity Market using Mixed Integer Non Linear Elsevier Materials Today: Proceedings, vol.5, pp.445-457, 2018. (Index in SCOPUS): <https://doi.org/10.1016/j.matpr.2017.11.104>
8. Manish Kumar, Ashwani Kumar, K.S. Sandhu, "Impact of distributed generation on nodal prices in hybrid electricity market", Elsevier Materials Today: Proceedings, vol.5, pp.830-840, 2018. <https://doi.org/10.1016/j.matpr.2017.11.154>
9. Manish Kumar, Karimula Polliseti, Ashwani Kumar, K.S. Sandhu, "Nodal Prices Determination with Wind Integration for Radial Distribution System", International Journal of Engineering, Science and Technology, vol.9,no.3,pp.11-21,2017. <https://doi.org/10.4314/ijest.v9i3.2>
10. M Kumar, A Kumar, KS Sandhu, "Reduced the Fuel Cost by Using Renewable Energy-Based DG in Pool Electricity Market", Recent Advances in Power Systems, LNEE, vol- 699, pp.415-42316, October 2020. https://doi.org/10.1007/978-981-15-7994-3_38 (Springer)
11. Manish Kumar, Nalin Chaudhary, "Saving of Fuel Cost by Using Wind+ PV-Based DG in Pool Electricity Market", Proceedings of International Conference on Trends in Computational and Cognitive Engineering, Advances in Intelligent Systems and Computing (AISC), vol- 1169, pp 293-30301, October 2020. https://doi.org/10.1007/978-981-15-5414-8_24 (Springer)
12. Muralidhar Nayak Bhukya, Manish Kumar, V. Chandra Jagan Mohan, "Design and Development of a Low-Cost Grid Connected Solar Inverter for Maximum Solar Power Utilization", Recent Advances in Power Electronics and Drives, LNEE vol-707, pp. 421-42904/12/2020,. https://doi.org/10.1007/978-981-15-8586-9_37 (Springer)
13. Manish Kumar, Muralidhar Nayak Bhukya, Anshuman, Sachin, "Impact and Scope of Electric Power Generation Demand Using Renewable Energy Resources Due to COVID-19", CoMSO, Smart Innovation, Systems and Technologies, vol-206, pp-495–502, 18/03/2021. DOI:10.1007 /978-981-15-9829-6_39 (Springer)
14. Manish Kumar, Muralidhar Nayak Bhukya, Anshuman, Sachin, "Escalating Demand, Present and Future Status on Hybrid Electric Vehicles", CoMSO, Smart Innovation, Systems and

- Technologies, vol-206, pp. 599-611, 18/03/2021 . DOI:10.1007 /978-981-15-9829-6_47 (Springer)
15. Muralidhar Nayak Bhukya, Manish Kumar, Shobha Rani Depuru, “A Simple Approach to Enhance the Performance of Traditional P&O Scheme Under Partial Shaded Condition by Employing Second Stage to the Existing Algorithm”, CoMSO, Smart Innovation, Systems and Technologies, vol 206, pp 545-556, 18/03/2021. DOI:10.1007 /978-981-15-9829-6_43 (Springer)
 16. Muralidhar Nayak Bhukya, Manish Kumar, Vipin, Chandervanshi, “Factors Affecting the Efficiency of Solar Cell and Technical Possible Solutions to Improve the Performance”, CoMSO, Smart Innovation, Systems and Technologies, vol 206, pp 623-634, 18/03/2021. DOI:10.1007/978-981-15-9829-6_49 (Springer)
 17. Bhukya, M.N., Sai, P.T.K., Kumar, M., Depuru, S.R., Babu, T.S. (2022). A Novel Hybrid GMPPT Scheme Based on P&O-MM with Reduced Output Power Oscillations Under PSC for PV System. Power Electronics and High Voltage in Smart Grid. Lecture Notes in Electrical Engineering, vol 817. Springer, Singapore. https://doi.org/10.1007/978-981-16-7393-1_24
 18. Kumar, P., Kumar, M., Bansal, A.K. (2022). Renewable Energy Resource Availability and Supply Guide in India. In: Kumar, A., Srivastava, S.C., Singh, S.N. (eds) Renewable Energy Towards Smart Grid. Lecture Notes in Electrical Engineering, vol 823. Springer, Singapore. https://doi.org/10.1007/978-981-16-7472-3_4
 19. Manish Kumar, Akshat Kant, Punit, Aakash and Ashu Singh, “ASSESSMENT OF POLICY AND POTENTIAL AVAILABILITY TO GENERATE ELECTRICITY USING WIND ENERGY “, International Journal of Mechanical Engineering, issue-4, vol.7, page-440-451, 2022. https://kalaharijournals.com/resources/APRIL_54.pdf
 20. Manish Kumar, Akshat Kant, Punit, Aakash and Ashu Singh, “Status and Utility of Solar Energy at Haryana in India “, International Journal of Mechanical Engineering, issue-4, vol.7, page-434-439, 2022. https://kalaharijournals.com/resources/APRIL_53.pdf
 21. Manish Kumar, Akshat Kant, Parveen Kumar, Ashwani Kumar and K. S. Sandhu, “HYDRO AND WIND PARTICIPATION FOR CONGESTION MANAGEMENT IN HYBRID ELECTRICITY MARKET”, Semiconductor Optoelectronics, Vol. 42 No. 1 (2023), 315-340. <https://bdtgd.cn/article/view/2023/315.php>
 22. Manish Kumar, Ashwani Kumar, K.S. Sandhu, “Optimal Location of PV based Distributed Generation in Pool based Electricity Market using Mixed Integer Non Linear Programming”, International Journal of Hybrid information Technology, vol. 9, no-10, pp-335-352, 2016. (Index in SCOPUS)
 23. Manish Kumar, Harish Kumar, Narveer “Impact of Wind Units in Congestion Management for Hybrid Electricity Market”, Proceedings of International Conference on Advancements in Computing & Management (ICACM) 2019. <http://dx.doi.org/10.2139/ssrn.3462577> (Index in SCOPUS)

Book Chapter (Index in SCOPUS)

1. Manish Kumar, Ashwani Kumar, K. S. Sandhu, “ WT based distributed generation location minimizing transmission loss using mixed integer non-linear programming in deregulated electrical market”, proceedings of the International congress on information and communication technology ICICT-2015, Advances in intelligent system and computing,vol.438, pp.443-358, 9-10 Oct,2015.
2. Manish Kumar, Ashwani Kumar, K. S. Sandhu, “ PV-based distributed generation location using mixed integer non-linear programming in deregulated electricity market”, Advanced computing & communication technology ICACCT 2015, Advances in intelligent system and computing,vol.452, pp.535-547,Nov, 2015.

International Journal

1. Manish Kumar, Manjeet , Pooja Khatri, “Study of faults on HVDC transmission lines”, Golden Research Thought, Volume3, Issue-8, Feb-2014.
2. Manish Kumar, “Transient Stability Enhancement of Two Area System using FACTS Controller”, Golden Research Thought, Volume3, Issue-9, March-2014.
3. Manish Kumar, “Comparative study and applications of FACTS devices in power system”, Indian Streams Research Journal ISSN 2230-7850) Volume-4 | Issue-2 | March-2014.
4. Manish Kumar, “Power quality improvements using DSTATCOM”, Research Directions Volume 1, Issue 10,PP-1-8, April 2014

International conference:

1. Ashwani Kumar, Manish Kumar and K. S. Sandhu, “Optimal DFIG Location and Impact of Load Model in Pool Electricity Market”, Proceedings of the 2014 International Conference on Power Systems, Energy, Environment (PSEE 2014), Interlaken, Switzerland, pp.149-156, Feb. 2014.
2. Manish Kumar, Ashwani Kumar, K. S. Sandhu, “Optimal Location and Sizing of Distributed Generation Sources considering Voltage Control Areas”, International Conference on Emerging Trends in Electrical, Electronics, Instrumentation& Computer Engineering (ETEICE), Bhilai Institute of Technology, Raipur & Institute for Research and Development India,Bhubaneswar, 27th march,2014.
3. Manish Kumar, K. S. Sandhu, Ashwani Kumar, “ Wind generation integration impact on fuel cost saving in pool based electricity market”, 5th international conference on advance in energy research (ICAER) held at IIT Bombay , 15-17 Dec 2015.
4. Manish Kumar, Manjeet, “Comparative analysis of STATCOM and UPFC device in power system”, International conference on advances in Computing & Communication Engineering 22-23 Feb.2014.
5. Manish kumar, K.S Sandhu, Ashwani Kumar, “Simulation analysis and THD Measurements of Integrated PV and Wind as Hybrid System Connected in Grid ”, 6th IEEE India International

Conference on Power Electronics(IICPE-2014), National Institute of Technology, Kurukshetra, 8-10 dec,2014.

6. Manish Kumar, Ashwani Kumar, K.S. Sandhu, “PV-WT based distribution generator location minimizing transmission loss in Pool/Bilateral electricity market model”, RAEREST 2016, Procedia Technology, vol.25, pp-692-701, 2016. Elsevier
7. Muralidhar Nayak Bhukya, Manish Kumar, Akshat Kant and Punit , “Renewable Energy: Potential, Status, Targets and Challenges in Rajasthan”, Journal of Physics: Conference Series 1854 (2021) 012004 doi:10.1088/1742-6596/1854/1/012004
8. Muralidhar Nayak Bhukya, Manish Kumar, Akshat Kant and Punit , “Renewable Energy: Potential, Status, Targets and Challenges in Rajasthan”, Journal of Physics: Conference Series 1854 (2021) 012004 doi:10.1088/1742-6596/1854/1/012004
9. Manish Kumar, Akshat Kant and Punit , “Environmentally friendly power: potential, status and challenges in Jharkhand”, 2021 International Conference on Design Innovations for 3Cs Compute Communicate Control (ICDI3C), page-135-139,IEEE. 27 September 2021. DOI: 10.1109/ICDI3C53598.2021.00036.
10. Manish Kumar, Akshat Kant and Punit , “Arduino Based system to prevent vehicle accidents”, 2021 International Conference on Design Innovations for 3Cs Compute Communicate Control (ICDI3C), page-140-148,IEEE, 27 September 2021. DOI: 10.1109/ICDI3C53598.2021.00037.
11. Manish Kumar etc al, “Photovoltaic (PV) Cell Parameter Objective Optimization Using Novel Convolutional Neural Network (CNN) Technique”, 2022 International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS),IEEE Explore, pp. 282-286, 2023, DOI: 10.1109/SSTEPS57475.2022.00076
12. Manish Kumar etc al, “Advanced Security for MANET using Ant Colony Optimization and Artificial Neural Network”, 2022 International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS),IEEE Explore, pp. 356-358, 2023, DOI: 10.1109/SSTEPS57475.2022.000793
13. Parveen Kumar, Manish Kumar, Ajay Kumar Bansal, “Implementation of Harris Hawk Optimizer for detection of Fault in PV module”, 2022 International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS),IEEE Explore, pp. 57-62, 2023, DOI: 10.1109/SSTEPS57475.2022.00027
14. Parveen Kumar, Manish Kumar, Ajay Kumar Bansal, “Monitoring of Grid-Connected Photovoltaic Systems using Multilayer Neural Networks”, 21/12/2023 9th International Conference on Signal Processing and Communication (ICSC), IEEE, pp. 380-386.
15. A. Kant, M. Kumar, and S. Sihag, “Robust behaviour of cobalt free LIBs for electric vehicle applications,” in 2025 7th International Conference on Energy, Power and Environment (ICEPE), 2025, pp. 1–5.

	16. A. Kant, M. Kumar, and S. Sihag, “Enhanced Battery Management in EVs Using Hybrid Duty Cycle Balancing for Optimal Battery Performance,” Accepted in 3rd International Conference on Smart Grid Energy Systems and Control (SGESC–2025).
Conference Organized:	- Organizing Secretary of International Conference on Smart and Sustainable Technologies in Energy and Power Sector (SSTEPS)-2022 “Organizing Publicity Chairs” International Conference on Future Power Network and Smart Energy Systems: Issues and Challenges (FPNSES-2023) during 28-30 Dec 2023 organized by Department of Electrical Engineering, NIT Kurukshetra, Haryana
Workshop Organized :	Three workshop organized as a coordinator : 5day workshop on MATLAB (8- 12th March,2019) 5 Day workshop on Arduino software (26-30th April 2019) 5Day workshop on Application of Optimization Techniques in power system using GAMS & MATLAB software.
Event Organized:	➤ Webinar Series on “Awareness on Alcoholic Addiction and Drug Abuse among Youths” under Nasha Mukth Samaj Abhiyan Andolan kaushal Ka Govt of India. ➤ V.C Trophy organized on 3rd May, 2019 as a Coordinator ➤ Induction Program of B.Tech 1st year students 2019 ➤ “Nasha Mukth Bhart Abhiyan” organized with Ministry of Social Justice and Empowerment , Govt. of India, 04/08/2022 ➤ “नशा मुक्त सामाजिक संकल्प समारोह” in DU Delhi,9/08/2022 ➤ रामायण की कहानी, विज्ञान की जुबानी एव नशामुक्त समाज आंदोलन अभियान कौशल का,10/08/2022 ➤ “Bhagwan Valmiki Jayanti – pargat Diwas”, 3/10/2022 ➤ Constitution Day, 21 November 2022. ➤ Motivational lecture for UPSC Aspirants, 4May 2023 ➤ “नशा मुक्त सामाजिक संकल्प समारोह” in Talkatora Stadium Delhi, 09/08/2023 ➤ Gandhi- A Saint with Multidimensional Personality, 1/10/2024 ➤ IEEE WIE Speech Competition, 6/05/2025
Labs Established :	- Control system Engineering Lab - Analog Electronic Lab - Microprocessor & interfacing Lab - Workshop Lab - Machine shop - Carpentry shop - Fitting shop - Welding shop - Foundry shop - Energy Lab-1
Workshop attended	More than 20 workshop attended